Renewable Energy 139 (2019) 739–752

Contents lists available at ScienceDirect

Renewable Energy

journal homepage: www.elsevier.com/locate/renene

Quantifying the revenue gain of operating a cascade hydropower plant system as a pumped-storage hydropower system

Mümtaz Ak [a], Elcin Kentel [b, *], Secil Savasaneril [c]

[a] Netsu Group Engineering and Consultancy Firm, Ankara, Turkey
[b] Middle East Technical University, Department of Civil Engineering, Ankara, Turkey
[c] Middle East Technical University, Department of Industrial Engineering, Ankara, Turkey

ARTICLE INFO

Article history:
Received 5 November 2018
Received in revised form
6 February 2019
Accepted 21 February 2019
Available online 28 February 2019

Keywords:
Pumped-storage hydropower systems
Operating rule-curve
Hourly electricity price variations

ABSTRACT

Pumped-storage hydropower is one of the most viable large-scale energy storage options. When managed optimally, pumped-storage hydropower may also bring monetary benefits due to price variations in the electricity market. Two main concerns are the lack of suitable sites and potential environmental impacts, which can be alleviated by using already existing, closely situated reservoirs. The goal of this study is to develop operating strategies for cascade pumped-storage hydropower systems composed of already existing hydropower plants. Main challenges are the treatment of the stochastic behavior of inflows and hourly electricity price variations. In this study, monthly historical inflows are used as inputs to generate operating rule curves. Based on past electricity prices, a scenario-based approach is developed to reflect the uncertainty in electricity price variations. Combining the two approaches, nonlinear mathematical models are constructed to obtain average annual revenues for cascade hydropower plants and the pumped-storage hydropower systems. The models are solved for the cascade multi-reservoir system in Coruh Basin of Turkey for five different scenarios based on electricity prices for years 2013–2017. The revenue gain ranged between 2.9% and 10.4%. It is concluded that operation of the cascade hydropower plant system in the pumped-storage mode brings additional revenue.

© 2019 Elsevier Ltd. All rights reserved.

1. Introduction

A power system is characterized by continuously varying loads due to fluctuating demand and supply, where the latter might be due to the uncertainties inherent in immediate availability of renewable energy sources such as wind, solar or hydro. To prevent the risk of mismatch in demand and supply on an hourly or daily basis, a power system needs storage schemes. Hydropower with reservoirs and pumped-storage are the most widely used forms of energy storage [1–4]. In addition to being domestic sources, they offer the ability to supply peak demands [5] and buffer the intermittent supply of electricity from renewable power sources such as wind and solar [6]. Pumped-storage hydropower (PSH) represents almost 99% of current worldwide electricity storage capacity [7]. PSH systems are used as energy storage mechanisms to shift availability of water to high-demand or high-benefit time zones. Although some energy is lost in the process, typically 15–25%, PSH system still contributes significantly by providing benefits to the grid in the form of peaking power, frequency stabilization, and load balancing [8]. In addition to these benefits, optimized operation strategies for PSH systems exploiting the fluctuations in electricity prices results in revenue gain.

Although beneficial, a major difficulty in constructing PSH system is the lack of suitable sites [4,9,10] and a major drawback is undesirable environmental impacts which can be reduced by utilizing the existing reservoirs [1]. When a PSH system is to be constructed on an existing pair of reservoirs, the topology should allow for pumping of water in between. Gimeno-Gutiérez and Lacal-Arántegui [7] assessed PSH potential in Europe, by considering pairs of already existing reservoirs with an adequate difference in elevation and which are situated close enough to each other. When a maximum of 20 km between existing reservoirs is considered, theoretical PSH potential for Turkey is estimated as 36.8 TWh while the total potential for EU is estimated only as 11.4 TWh [7]. As an energy importing country with limited thermal energy sources, Turkey has 16% of the economically viable hydropower potential of Europe [11] and evaluation of PSH system possibilities together

* Corresponding author.
E-mail addresses: akmumtaz@hotmail.com (M. Ak), ekentel@metu.edu.tr (E. Kentel), ssecil@metu.edu.tr (S. Savasaneril).

https://doi.org/10.1016/j.renene.2019.02.118
0960-1481/© 2019 Elsevier Ltd. All rights reserved.

with hydropower is worth investigating. There are no PSH systems in Turkey yet, and selection of already existing cascade HPP systems to be converted to PSH systems is a challenge.

In our work, we consider a multi-reservoir cascade hydropower system where all reservoirs are designed and planned for conventional hydropower generation. The PSH system uses the reservoirs of the conventional cascade system as upper- and lower-reservoirs. Transforming the existing HPP to a PSH system requires mainly replacing the turbines of all reservoirs other than the most downstream one with reversible turbines. A reversible turbine is a machine that can operate in three modes of operation, in pumping mode, in turbine mode and at idle speed. In the turbine mode, water stored in the upstream reservoir is released to the downstream reservoir to produce electricity; while in the pump mode, electricity is used to pump water from the downstream reservoir to the upstream one. The main purpose is to shift availability of water to high-demand or high-benefit time zones. Such a PSH system, where both the upper and lower reservoirs are fed by inflows is called an open-loop PSH system.

Developing operating rule curves for open-loop PSH systems is challenging due to the stochastic nature of both electricity prices and inflows, and due to the nonlinear relationships between the elements of the reservoir (such as head and volume, or the energy generated). Although impact of electricity price variations on the revenue of PSH systems have been investigated in the literature, simultaneous treatment of inflow and electricity price variability is rare. For example, Connolly et al. [12] investigated the possibility of profiting from a pumped hydroelectric energy storage facility on existing markets and concluded that even with low investment costs, a low interest rate, and a suitable electricity market; they are risky investments without a predictable profit. Notwithstanding, they stated that PSH systems will have a large role in future electricity grids. Wilson et al. [13] highlighted the challenges in forecasting the time-shifting revenues for PSH systems and variability in the level of revenue from year to year. In their study, Wilson et al. [13] focused on the potential storage revenues based on the actual electricity prices for a typical PSH system. However, explicit evaluation of annual operating policies on revenue gain due to hourly electricity price variations are neither presented in these or other similar studies [14–16]. provided an extended literature review on PSH and concluded that almost all the studies were devoted to formulating and solving short-term optimization problems, suggesting that monthly inflow variations are ignored. For instance, Pérez-Díaz et al. [17], similar to our study, consider two reservoirs in cascade form on a river, where the lower reservoir is a re-regulation reservoir and the water is pumped to the upper reservoir. Electricity prices of an earlier year is used to quantify the (net) revenue. However, the MIP model developed in that study is only for a planning horizon of one month. Bregar [18] studies three cascade reservoirs on Soca river. A pumped-storage reservoir is build as an upper reservoir using the reservoir in the middle as the lower reservoir. The benefit of PSH is measured taking the planning horizon as one day. Hunt et al. [19] analyzes the energy benefit of building an artificial site for pumped-storage in Amazon region, which will help to store water during wet season to generate energy during the dry season. In Pérez-Díaz et al. [20] it is stated that introduction of electricity market considerations in the operation strategy models that have long time horizons still today remains a challenging task.

Fleten et al. [21] stated that the presence of reservoirs means that the short-term scheduling problem of determining bids for the next 12–36 h is a part of a long-term problem in which the question is whether to release water now or store it for the future. To address this multi-scale challenge, variations of the following approaches are proposed in the literature: (i) to use several models for hydropower planning, at least one long-term model and one short-term model (for example [21]), (ii) to decompose the problem into a planning hierarchy where the long/mid-term models provide boundary conditions for the short-term models (for example [22]), (iii) to couple short- and long-term decisions into the same model (for example [16,25] and this study). Tica et al. [16] developed a methodology for the performance estimation of a PSH system at the annual level, using the inflows from the mean hydrological year. Our study differs from Tica et al. [16] such that we use 19 years of past inflow to treat yearly inflow variations instead of a mean hydrological year in evaluating the average annual revenue. Bozorg-Haddad et al. [25] quantifies the benefit of converting a conventional HPP to a PSH for a two-reservoir cascade system located on a river. The electricity prices are approximated with two constant values (at high and low levels) throughout the simulation horizon. The past inflow data is taken as the input. We differ from Bozorg-Haddad et al. [25] in two important ways. Firstly, we use all past information about inflows while developing rule curves that could be used for the upcoming year. Secondly, we use the exact hourly year-long price information to estimate the benefit and to precisely determine monthly energy generation.

The coupling of short- and long-term decisions into a single model where one stage (for example a month during which an optimum amount of water release is realized) may represent many trading periods (for example hours for which electricity prices vary), requires an attempt to preserve some representation of the price variability within a stage [23]. In other words, the monthly dispatch amounts require an aggregate approach to incorporate the day-ahead prices. In this study, we address the challenge of deriving an operating strategy for a long planning horizon (such as one to several years), that simultaneously considers the variations in monthly inflows and the variations in hourly price fluctuations in a novel way (discussed in details in Section 2). Our aim in deriving an operating strategy is to come up with an estimate of the revenue gained from operating a HPP or PSH for a yearlong period.

Another challenge in developing rule curves for multi-reservoir cascade PSH systems is the increased number of decision variables since for each reservoir optimal allocation of the available time between turbining and pumping needs to be determined. Common approaches for deriving rule curves for multi-reservoir systems include linear programming [26,27], nonlinear programming [28,29], soft computing algorithms [30–32], and dynamic programming [33,34]. Recently, approaches such as orthogonal progressive optimality algorithm (OPOA) [35] and adopting dynamic programming in a supercomputing cluster [36] have been proposed to address the dimensionality problem. In this study, we developed nonlinear programming models for a cascade system composed of three hydropower plants. Therefore the curse of dimensionality is not as pronounced as for the cases where large number of hydropower plants are considered simultaneously.

The goal of this study is to evaluate the potential revenue gain of operating an existing HPP system as a PSH system. To quantify the potential gain, the revenues that can be generated under the existing HPP system and the envisioned PSH system are compared. Under both systems, optimum operating strategies for the integrated reservoir system are developed so that a fair comparison of maximum possible revenues would be made. The operating strategies that maximize revenues are characterized by operating rule curves, which identify end-of-month reservoir storages for each of the reservoirs.

Through this study we aim to make several major contributions. Our *first* contribution is to develop a method that enables the decision-makers to quantify how much a PSH system would increase the revenue of a conventional HPP. The developed method is

versatile and universal in that it is possible to use it for assessment of cascade reservoir systems, given that the topology is suitable, i.e., reservoirs are located sufficiently close (as discussed in Ref. [7]). Our *second* contribution is to develop a mathematical model to generate operating rule curves for cascade reservoir systems (located on rivers) for a conventional HPP and for a PSH system. The rule curves provide guidance on the long-term decisions of the amount of monthly discharges for the upcoming year, while taking into consideration the hourly prices within each month. All past monthly inflow information is utilized in the model, thus the stochasticity in inflows is incorporated implicitly. Our *third* contribution is to the practice of operating a cascade system. Through this study we gain insights as to how to operate the cascade reservoir system in Çoruh Basin, Turkey, under a conventional HPP and under a PSH system. Analysis shows that the position of a reservoir with respect to the others in the cascade system have a significant impact on the rule curves under both HPP and PSH systems. Furthermore, the long-term decision of monthly discharges is not much affected by the system being a HPP or PSH, whereas short-term decisions are predominantly affected. Those insights would be useful for the policy-makers as well as for the operators of the cascade systems.

2. The framework for analysis

We first construct the mathematical models for each of the operating modes, namely, the conventional HPP and the PSH system. The model takes two major inputs: the monthly inflows and the hourly day-ahead prices. The planning horizon is one year (the upcoming year) and the model yields a rule curve (for the upcoming year) in the form of end-of-month reservoir storage levels for each of the 12 months, for each of the reservoirs. To be able to reflect the monthly variations in inflows, the model explicitly makes use of the monthly inflows for 19 years of simulation period (a total of 228 data points for monthly inflows). Thus the information regarding the past inflow data is fully used in our model whereas the rule curves derived can be used in the future.

The second major input the model takes are hourly day-ahead price data for a yearlong period (a total of 8760 hourly data points for each year). We use the past realized day-ahead hourly market clearing prices for a period of one year. Note that day-ahead prices are determined dynamically daily through the interaction of many players in the energy market, it is neither possible nor meaningful to try to forecast the day-ahead hourly prices for a yearlong period. Rather than forecasting the prices for the future, in our revenue gain analysis we use the past realized prices. To incorporate the effect of price uncertainty on the estimate of the revenues, we consider several possible price-scenarios (i.e. developed based on hourly electricity prices realized in previous years). Each scenario consists of the electricity prices throughout the 24-hr day for the corresponding year (a different year for each of the five scenarios).

For each mode of operation (conventional or PSH), the mathematical model is run for each of the price scenarios. The mathematical models yield rule curves (based on end-of-month storage levels), total number of operation hours for each of the 12 months, and the maximum annual revenue that can be gained under the corresponding operating mode. The revenue value is obtained for each scenario and for each mode. Then the difference in the revenues (across operating modes) is calculated. The difference value provides an estimate of the additional revenue gain due to PSH. Combining the scenarios, we come up with a range of estimate of additional revenue gain.

3. Obtaining operating policies for hydropower plant and pumped-storage systems

In this section, we model (through a mathematical program) the cascade multi-reservoir system in the absence and in the presence of pumped-storage (referred to as HPP system and PSH system, respectively). Each model decides on how much water to turbine or pump every month to maximize the annual revenue. Then using data from our case study area (Section 4) and the realized electricity prices, we compare the two revenues to quantify the revenue gain of the PSH system in Section 5. In the PSH system, both turbining and pumping decisions are made, whereas in HPP only turbining decisions are made. Thus, the HPP system can be considered as a special case of the PSH system.

The PSH model takes monthly inflows and hourly electricity prices as inputs and yields an operating rule curve (i.e., end-of-month storage levels) and maximized revenue, as the outputs. The operating rule curve will guide the operator for the upcoming water year in terms of discharge amounts for turbining and pumping to maximize the revenue. Both monthly inflows and electricity prices are stochastic in nature and we tackle with stochasticity as follows. We use all available historical inflow data for the past n years in the model (in our case for the past 19 years, see Section 4) and the model gives the operating rule curve for each reservoir. Since the operating rule curves are obtained taking into consideration all past realized inflows, it is expected to operate well, given that the future inflows will be similar to those occurred in the past.

It is assumed that the system under consideration is a small company who acts as a price-taker in the electricity market (see Refs. [24,25,28,37–39], for a similar assumption). This implies the hourly day-ahead prices are taken as the prices that the electricity can be sold at. In Turkey, all the electricity generated is sold to the grid and all the demand is supplied from the grid. Therefore, impact of individual HPPs in terms of responding to the load change of the power system cannot be explicitly identified. However, hourly electricity prices inherently includes impact of load variations within the power system. Whenever the demand increases and the supply is limited, the electricity prices increase and vice versa. The mathematical model we developed in this study indirectly considers the effect of load variations by making use of hourly electricity prices.

To incorporate the stochasticity in the hourly electricity prices we follow a scenario-based approach. Using the past hourly price data for different scenarios (where each year corresponds to a different scenario) we run the model and obtain a range for possible average annual revenues. Note that our model gives an estimate of average annual revenue, and it is aggregate in that rather than hourly discharge amounts, monthly discharge amounts are decided on. Thus, it is meaningful to take a scenario-based approach. The flowchart in Fig. 1 shows the inputs and the outputs of the model more explicitly. In Section 3.1 we present the mathematical model for PSH, and then in Section 3.2 the changes made in the model to operate the system as a HPP are presented. In Section 3.3 we explain how hourly electricity prices are aggregated to calculate monthly turbining revenues and pumping costs.

3.1. Mathematical model for the pumped-storage hydropower system

In this section, first, the mathematical model is introduced together with the notation. Then the objective function, which is the maximization of the average annual energy revenue, *AAER* and the constraints are described in details.

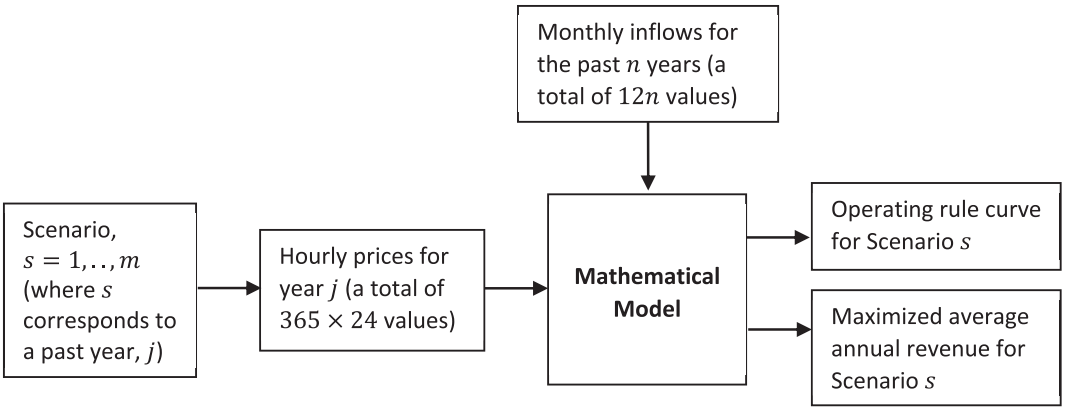


Fig. 1. Flow chart for the mathematical model.

$$Max.\ AAER = \frac{1}{n}\sum_{i=1}^{i=K}\sum_{t=1}^{T}\Big[E_{t,i} \times P_t(Tm_{t,i}) - EPump_{t,i} \times PPump_t\Big(Tmpump_{t,i}\Big)\Big] \quad (1)$$

$$E_{t,i} = (H_{t,i} - hloss(Q_{t,i})) \times Q_{t,i} \times 30 \times Tm_{t,i} \times \mu_i(Q_{t,i}) \times \gamma,\ \ t\in T,\ i\in K \quad (2)$$

$$EPump_{t,i} = \Big(H_{t,i} + hloss\Big(Qpump_{t,i}\Big)\Big) \times Qpump_{t,i} \times 30 \times Tmpump_{t,i} \times \frac{\gamma}{\mu p_i\Big(Qpump_{t,i}\Big)},\ \ t\in T,\ i\in K \quad (3)$$

$$H_{t,i} = f_i\left(\frac{S_{t-1,i} + S_{t,i}}{2}\right) - tail_i,\ \ t\in T,\ i\in K \quad (4)$$

$$Tm_{,ti} = \frac{\min(O_{t,i},\ Omax_i)}{3600 \times 30 \times 10^{-6} \times Q_{t,i}},\ \ t\in T,\ i\in K \quad (5a)$$

$$Tmpump_{t,i} = \frac{\min\Big(OPump_{t,i},\ Omax_i\Big)}{3600 \times 30 \times 10^{-6} \times Qpump_{t,i}},\ \ t\in T,\ i\in K \quad (5b)$$

$$Tmpump_{t,i} + Tm_{t,i} \le 24,\ \ t\in T,\ i\in K \quad (6)$$

$$IA_{,ti} = R_{t,i-1} + O_{t,i-1} + I_{t,i} - I_{t,i-1},\ \ t\in T,\ i>1 \quad (7a)$$

$$IA_{t,i} = I_{t,i},\ \ t\in T,\ i=1 \quad (7b)$$

$$S_{t,i} = S_{t-1,i} + IA_{t,i} - R_{t,i} - O_{t,i} - Opump_{t,i-1} + Opump_{t,i},\ \ t>1,\ i>1 \quad (8a)$$

$$S_{t,i} = S_{t-1,i} + IA_{t,i} - R_{t,i} - O_{t,i} + Opump_{t,i},\ \ t>1,\ i=1 \quad (8b)$$

$$Smin_i \le S_{t,i} \le Smax_i,\ \ t\in T,\ i\in K \quad (9)$$

$$R_{t,i} \ge Rw_i,\ \ t\in T,\ i\in K \quad (10)$$

$$S_{0,i} = S_{T,i},\ \ i\in K \quad (11)$$

$$Qmin_i \le Q_{t,i} \le Qd_i,\ \ t\in T,\ i\in K \quad (12a)$$

$$Qmin_i \le Qpump_{t,i} \le Qd_i,\ \ t\in T,\ i\in K \quad (12b)$$

$$S_{t,i} = S_{t+12,i},\ 0<t<217,\ \ i\in K \quad (13)$$

where.

Sets:
t (month): Time step ($t\in T = \{1,2,3,\ldots,12\times n\}$)
i (Plant): Hydropower plant ($i\in K = \{1,2,..,I\}$)
Scalars:
$Smin_i$: Minimum level of reservoir storage (hm^3) for HPP i
$Smax_i$: Maximum level of reservoir storage for (hm^3) HPP i
S_{0i}: Initial reservoir storage (hm^3) for HPP i
Rw_i: Minimum water release as environmental flow (hm^3) for HPP i
$Qmin_i$: Minimum turbine discharge (m^3/s) for HPP i
Qd_i: The design discharge for turbining or pumping (m^3/s) of HPP i
γ: is the specific weight of the water (9.81 kN/m^3)
n: The number of years in the operation period, $T/12$
$tail_i$: tail water elevation (m) for HPP i
Kl_i , Kp_i: coefficients used in $hloss(q)$ function (defined below), for the head losses that occur through the tunnel and the penstock, respectively
Dl_i , Dp_i: the diameters (m) of the tunnel and the penstock, respectively
Parameters:
I_{ti}: Water volume (hm^3) entering the reservoir naturally for power plant i in month t

(*continued*)

$f_i(s)$: Head level (m) given that s is the volume for reservoir of HPP i, where s is in hm^3 (see Section 3)

$hloss(q)$: Defined in meters as $hloss(q) = Kl_i\frac{\left(\frac{q}{\pi Dl_i^2/4}\right)^2}{2g} + Kp_i\frac{\left(\frac{q}{\pi Dp_i^2/4}\right)^2}{2g}$, q denotes the discharge (m^3/s), Kl_i , Kp_i, and Dl_i , Dp_i are defined as above

$\mu_i(y)$: efficiency of the **turbine** in HPP i, given the ratio, y, of discharge q (m^3/s) to Qd_i (a function of $y = \frac{q}{Qd_i}$)

$\mu p_i(y)$: efficiency of the **pump** in HPP i, given the ratio, y, of pump discharge q (m^3/s) to Qd_i (q and Qd_i are in m^3/s)

$P_t(x)$: representative price of electricity (TL/kWh), for **turbined** water in month t, given that duration of energy generation per day is x (a function of x).

$PPump_t(x)$: representative cost of electricity (TL/kWh), for **pumped** water in month t, given that duration of energy consumption per day is x (a function of x)

Variables:

$E_{t,i}$: the energy generated (kWh) at HPP i due to **turbining** in month t

$EPump_{t,i}$: the energy consumed (kWh) at HPP i due to **pumping** in month t

$Tm_{t,i}$: duration (hours) of energy generation for **turbining** in month t for HPP i

$Tmpump_{t,i}$: duration (hours) of energy generation for **pumping** in month t for HPP i

$H_{t,i}$: gross head (m) in month t for HPP i

$IA_{t,i}$: Actual water volume entering the reservoir (hm^3) for power plant i in month t

$S_{t,i}$: The reservoir water volume (hm^3) at the end of month t for HPP i, $S_{0,i} = 0$.

$Savg_{t,i}$: The average reservoir storage (hm^3) for month t for HPP i

$O_{t,i}$: The amount of water **turbined** (hm^3) in month t (sum of spilled water and water used for energy generation) for HPP i, $O_{t,i-1} = 0$ for $i = 1$.

$Opump_{t,i}$: The amount of water **pumped** (hm^3) in month t at HPP i (from reservoir of HPP $i+1$ to that of i), $Opump_{t,i-1} = 0$ for $i = 1$

$Q_{t,i}$: Discharge (m^3/s) passing through the **turbines** for energy generation in month t for HPP i

$Qpump_{t,i}$: Discharge (m^3/s) passing through the **pump** for energy generation in month t for HPP i

$R_{t,i}$: The water released (hm^3) as environmental flow in month t for HPP i

Let us first describe the objective function. In Equation (1), the net revenue is obtained as the difference of revenue gained due to energy generated, $E_{t,i}$, and revenue lost due to energy consumed, $EPump_{t,i}$, during pumping. For HPP i, the energy generated in month t, $E_{t,i}$, is obtained in Equation (2) as multiplication of net head, $H_{t,i} - hloss(Q_{t,i})$ and the amount turbined in month t at HPP i ($Q_{t,i} \times Tm_{t,i}$). This amount is multiplied with the efficiency factor and converted to kWh. The efficiency of the turbine at HPP i in month t, μ_i, is a function of $\frac{Q_{t,i}}{Qd_i}$, and is obtained from a specific energy curve of a Francis type turbine. Note that we used the same efficiency curve for pumping as well. The energy amount generated is then multiplied with the corresponding representative price, $P_t(Tm_{t,i})$, to obtain the revenue. In Section 3.3, we describe how representative price functions and energy revenue are obtained. Similarly, energy consumed for HPP i in month t, $EPump_{t,i}$, is obtained in Equation (3) considering the monthly amount pumped. Note in $EPump_{t,i}$ that net head is computed as $H_{t,i} + hloss(Qpump_{t,i})$, and the amount is divided by the efficiency factor. The energy consumed is multiplied with the corresponding representative cost, $Ppump_t(Tmp_{t,i})$ to obtain the energy expense.

Constraint given in Equation (4) expresses the net head using the head-volume relationship. In the case study, a second-degree polynomial equation is used to express this relationship (see Section 4.1, Figs. 5–7). To account for the beginning and the end-of period reservoir levels, the head is related to the average reservoir level for month t.

Constraint given in Equation (5a) gives the daily turbine duration in month t for HPP i. This duration is obtained as 1/30 of the monthly duration, where monthly duration depends on the outflow amount, $O_{t,i}$. Daily turbine duration cannot exceed 24 h (which is the maximum available hours per day), which is ensured by the parameter $Omax_i$. A similar expression gives the pumping duration in Equation (5b). Constraint given in Equation (6) ensures the total number of hours allocated to turbining and pumping cannot exceed 24 h per day. Constraints given in Equation (7a-b) define the water volume entering the reservoir of HPP i at time t, denoted with $IA_{t,i}$. Note that actual inflow depends on natural inflow as well as the outflow from the upper HPP. Equation (8a-b) are continuity equations. Note continuity should take into account the pumped water from the reservoir of HPP i to that of $i-1$ as a negative flow, and the pumped water from the reservoir of HPP $i+1$ to that of i as a positive flow. Equation (9) limits the end-of-month reservoir storages in accordance with the physical limitations of the reservoir. According to this constraint, neither water level nor water volume can exceed the corresponding minimum and the maximum values. Equation (10) sets a lower bound on the released water volume for environmental requirements, and Equation (11) forces the initial and final storage levels of the simulation period to be the same (initial reservoir levels are assumed to be at their maximum levels). Equation (12a-b) are used to limit the discharge rate for turbining and pumping, and allows the rates to differ from the design discharge. Finally, Equation (13) is used for the derivation of the operating rule curve (i.e. best end-of-month reservoir storages). This constraint ensures that best end-of-month reservoir storage values are obtained for the whole simulation period of n years, and that the operating rule curve can be used as a guidance in the upcoming years.

Next, we discuss how Equations 1–13 is modified to come up with a mathematical model for HPP. Then, in Section 3.3, we describe how representative price and cost functions $P_t(x)$ and $Ppump_t(x)$, are obtained as a function of turbining duration ($Tm_{t,i}$) or pumping duration ($Tmpump_{t,i}$), and how revenue and energy expense are calculated.

3.2. Mathematical model for the conventional hydropower system

The PSH system described in Section 3.1 is built up on HPP by allowing for the pumping of the water between reservoirs, besides turbining. Since in HPP only turbining of water is allowed and pumping is not allowed, HPP can be regarded as a special case of PSH. The same model derived in Equations 1–13 can be used for HPP, except that the variables related to pumping are set to zero. Indeed, setting $Tmpump_{t,i} = 0$, ensures that no pumping takes place and system operates as an HPP.

3.3. Representative price and cost functions

The representative price of electricity for turbined water for each month is obtained through the Variable Price Method (VPM);

for details see Ref. [28] or [39]. In Ak et al. [28], VPM is used to calculate the revenue for energy generation. Here, we use VPM for calculating both the energy revenue obtained from turbining and the energy cost due to pumping. We first describe the calculation of revenue.

It is assumed that, the total amount of water turbined in month t will be turbined in equal daily amounts. Every day, the turbining (i.e., energy generation) is assumed to take place during the hours in which the electricity prices are the highest, to maximize the revenue from energy generation. Such an approach assumes that generation of electricity is prioritized based on its price.

To calculate the revenue for a given month, first a *representative day* is constructed for that month, taking the average of prices corresponding to each hour throughout the month. As a result, a price vector for 24 h is obtained for that month. Then, based on the daily turbining duration (of that month), the corresponding highest prices of the price vector will be averaged, and multiplied with the daily energy generated. This will give daily revenue for that month, and multiplying with the number of days (30) will give the monthly revenue gained in that month.

We defined $P_t\ (x)$, as the *representative price function* for month t. This function takes the 24-h price vector for month t and the daily turbining duration x, and yields the average of x highest prices of the price vector. Considering the 24-h price vector, the values that x can take are 1,2,..,24, and thus 24 representative prices are available. However, since the monthly outflow values may indicate non-integer daily turbining durations, a second-degree function is fitted to the representative prices to obtain a continuous representative price function.

For example, for a given month, suppose the outflow amount indicates that water will be turbined for 150 h for that month. Then daily turbining duration would be $150/30 = 5$ h, and the electricity is generated during the 5 h in which the electricity prices are the highest. So, $P_t\ (5)$ would yield the average of maximum 5 electricity prices. If on the other hand the water will be turbined a total of 595 h for that month, daily turbining duration is 19.8, and $P_t\ (19.8)$ would yield the representative price of electricity. Here $P_t\ (19.8)$ is obtained by fitting a second-degree curve that passes through the discrete points $P_t\ (1),\ P_t\ (2), \ldots P_t\ (24)$. Note in the model in Equation (5a), the daily turbining duration is indicated with Tm_{ti}. For instance, considering the prices for year 2013, for month 5, for daily turbining duration x, $P_t\ (x)$ is obtained as (see Fig. 2):

$$P_5(x) = -0.031x^2 - 0.045x + 165.06 \quad 1 \le x \le 24 \qquad (14a)$$

As an example, the representative price functions for each month of year 2013, $P_t(x)$ is given in Fig. 2. In the numerical analysis, we use price data for years 2013, 2014, 2015, 2016 and 2017. Each year is treated as a distinct scenario and curves similar to those given in Fig. 2 are derived for other four years.

We next describe the calculation of energy expense due to pumping. It is again assumed that, the total amount of water turbined in month t will be turbined in equal daily amounts. However, now the pumping, i.e. energy consumption, is to take place during the hours in which the electricity prices are the lowest, to minimize the cost due to energy expense.

We defined, $Ppump_t\ (x)$, as the *representative cost function* for pumping for month t. This function takes the 24-h price vector for month t and the daily turbining duration x, and yields the average of x **lowest** prices of the price vector. Again a function is fitted to represent $Ppump_t\ (x)$ as a continuous function of x, daily pumping duration. Note in the model in Equation (5b) the daily pumping duration is indicated with $Tmpump_{t,i}$. For instance, considering the prices for 2013, for month 5, $Ppump_t\ (x)$ is obtained as (see Fig. 3):

$$Ppump_5(x) = -0.027x^2 + 3.143x + 77.628 \quad 1 \le x \le 24 \qquad (14b)$$

As an example, the representative price functions for each month of year 2013, $Ppump_t(x)$ is given in Fig. 3. In the numerical analysis since each year is treated as a distinct scenario similar curves are derived for other four years.

The models derived in Sections 3.1 and 3.2 are applied to data obtained from a cascade three-reservoir system in Çoruh Basin. Before comparing the outcomes of the models, in Section 4 we provide information on the case study area, and the other parameters used in the model.

4. Study area

Arkun, Yusufeli and Artvin are three consecutive HPPs in the middle basin of Coruh River located in the Northeast of Turkey. In Fig. 4, the locations of dams are shown and the characteristics of dams are given in Table 1.

The monthly inflow data from 1993 to 2011 are obtained from the General Directorate of State Hydraulic Works of Turkey [11] and used in the mathematical model in deriving the rule curves. Maximum flows and maximum spreads are observed in April, May and June. Statistical characteristics of monthly inflows for 19 years of simulation period for Arkun, Yusufeli and Artvin dams are provided together with the corresponding operating rule curves of the reservoirs in Section 5.2.

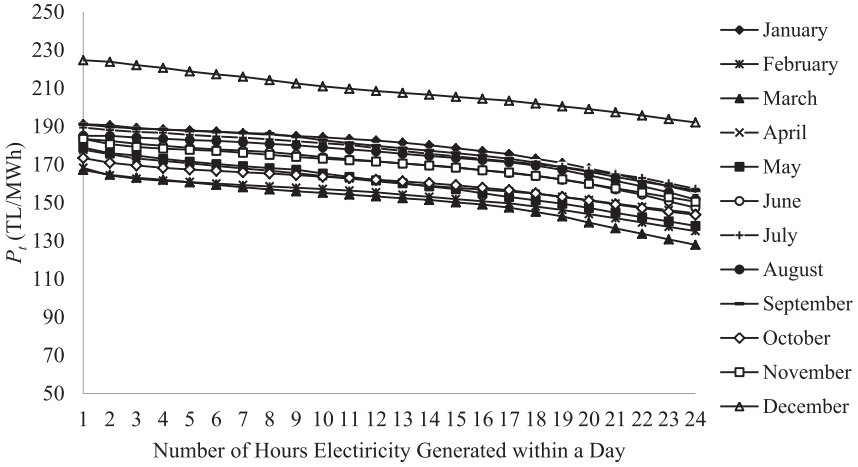


Fig. 2. Representative prices with respect to the duration of electricity generation in a day for 2013 [40].

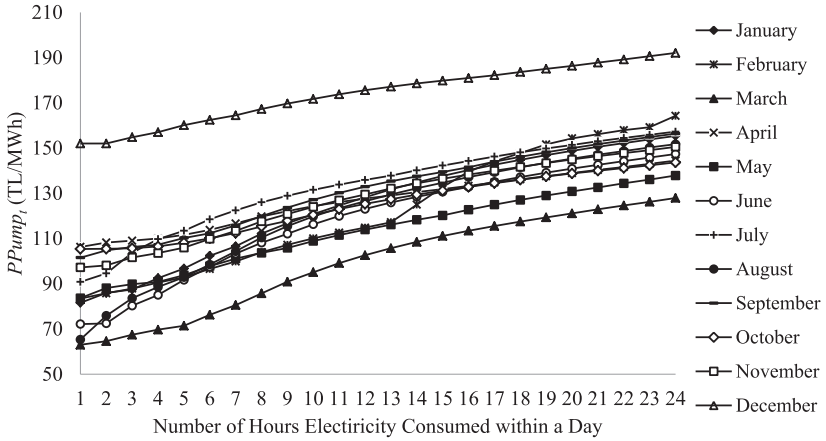


Fig. 3. Representative unit costs with respect to the duration of electricity consumption in a day for year 2013 [40].

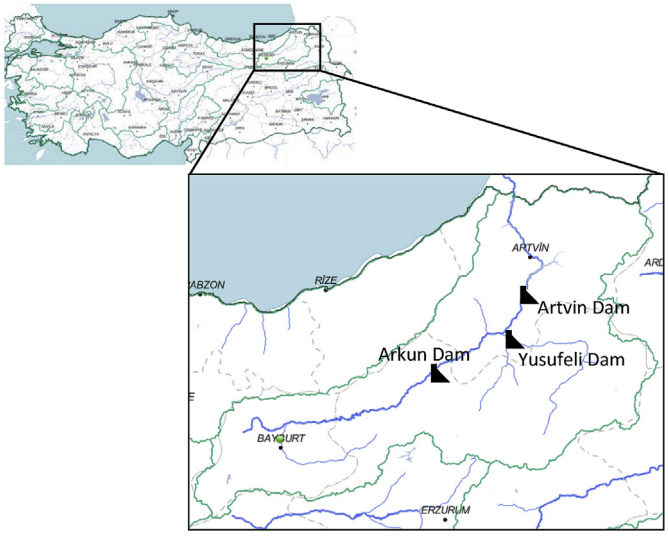


Fig. 4. Coruh river basin (Map is taken from rasatlar. dsi.gov.tr).

4.1. Topographical structure

The topographical data is used to derive the relation between head and storage (volume) for each reservoir. The estimated elevation-storage curves are given in Figs. 5–7, for Arkun, Yusufeli and Artvin, respectively. In these figures, $Savg_t = \frac{(S_t+S_{t+1})}{2}$.

5. Discussion of the results

The cascade three-reservoir system consisting of Arkun, Yusufeli and Artvin reservoirs is an open-loop one; the inflows already allow for revenue generation in conventional HPP. In this section, we quantify the amount of additional revenue generated due to PSH System. In the PSH System, three reservoirs are assumed to be connected through pumps, which allow transfer of water from downstream reservoirs to upstream reservoirs in addition to conventional turbines used for electricity generation. The following questions are addressed: Firstly, is there a significant revenue potential in the PSH system beyond the conventional system (HPP)? Secondly, how do the operating rule curves change? Finally, how do conversion into a PSH system affects the utilization of turbines and the total energy generated? These questions are addressed by solving the nonlinear programming models constructed in Sections 3.1 and 3.2. GAMS.24.2.3 is used for programming purposes and

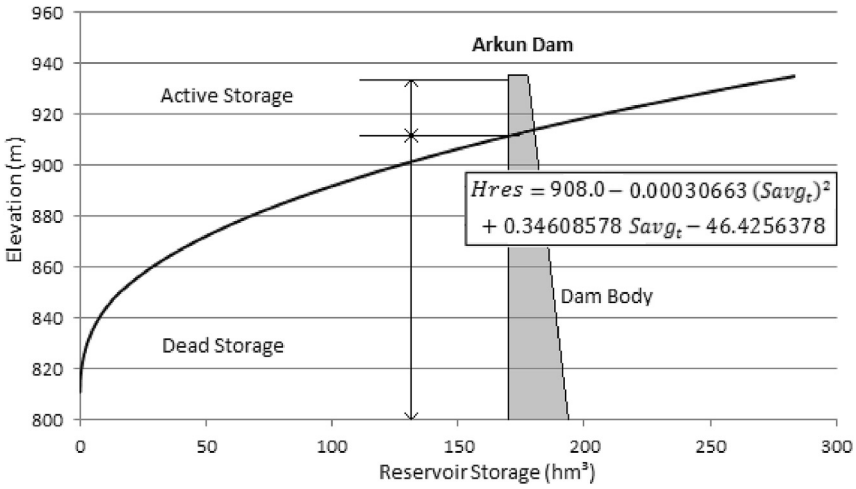


Fig. 5. The volume-elevation curve of Arkun HPP reservoir, $f_1(S_{avg})$ in Equation (4).

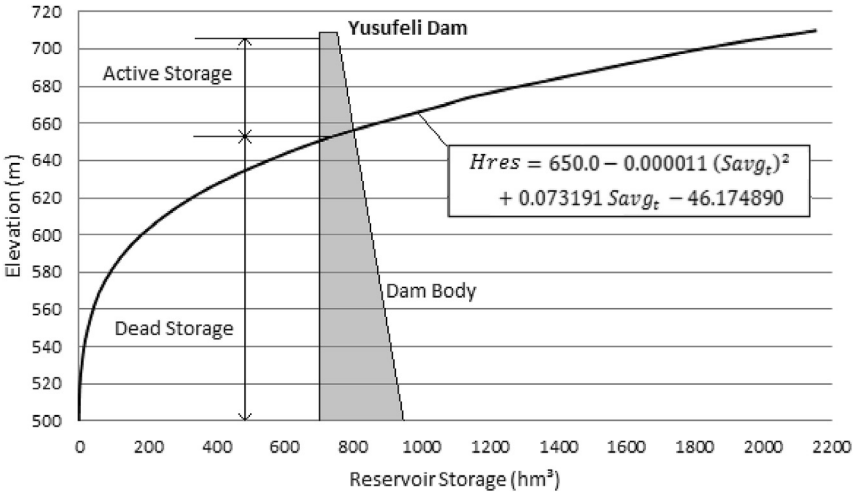


Fig. 6. The volume-elevation curve of Yusufeli HPP reservoir, $f_2(S_{avg})$ in Equation (4).

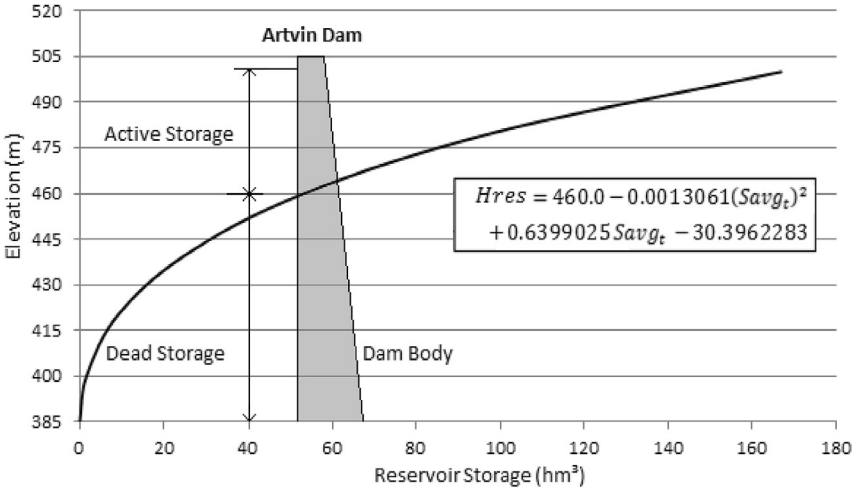


Fig. 7. The volume-elevation curve of Artvin HPP reservoir, $f_3(S_{avg})$ in Equation (4).

Table 1
The characteristics of arkun, yusufeli and artvin dams [41].

Characteristics	Arkun	Yusufeli	Artvin
Purpose	Energy	Energy	Energy
Average Annual Flow (hm^3)	1860.70	3684.24	3754.32
Dead Storage Volume (hm^3)	163.65	692.70	98.60
Maximum Storage Volume (hm^3)	283.24	2156.68	166.80
Active Storage Volume (hm^3)	119.59	1463.98	68.20
Reservoir Area (km^3)	5.53	33.21	15.43
Design Discharge (m^3/s)	124.98	321.00	160.89
Installed Capacity (MW)	238.00	540.00	332.00
Gross Head (m)	225.86	208.90	116.20
Net Head (m)	222.65	199.68	107.21

CONOPT is used as the solver. The solutions are obtained in around 60 CPU time.

5.1. Quantifying the additional revenue of pumped-storage hydropower over conventional system

In this study, the mathematical model is solved for five different scenarios, representing years between 2013 through 2017. In all scenarios, monthly inflows of past 19 years is used. Therefore, total 10 (one for HPP system and one for PSH system for each year between 2013 and 2017) mathematical models are solved. As a result, 10 rule curves and 10 average annual revenues are obtained. Comparison of revenues obtained for HPP and PSH systems are summarized in Table 2. For the PSH System, revenue of turbining, cost of pumping and the net revenue, which is the difference of these two, are reported.

When the revenues of turbining for HPP and PSH systems are compared, it is observed that the gain ranges between 22.7% and 31.9%. However, under the PSH system there is a cost associated with energy consumption during pumping, which does not exist in the conventional system. When the energy cost of pumping is subtracted, the net revenue of the PSH System is obtained (see Fig. 8). As can be seen from the last column of Table 2, percent gain in net revenue due to operating the HPP system as a PSH system ranges between 2.9% and 10.4% and has an average of 6.0%. Distribution of the revenue among three reservoirs for HPP and PSH systems is given in Fig. 9.

As can be seen from Fig. 9, the net revenue generation for Artvin reservoir (the most downstream reservoir) decreased while those of Arkun and Yusufeli reservoirs increased when the HPP system is

Table 2
Revenues obtained for HPP and PSH systems for each scenario.

Scenario		HPP System	PSH System			% Gain in revenue of turbining	% Gain in net revenue
No	Year	Revenue of turbining[a] (10^6 TL/year)	Revenue of turbining (10^6 TL/year)	Cost of pumping (10^6 TL/year)	Net revenue[b] (10^6 TL/year)		
1	2013	613.95	790.14	139.76	650.38	28.7	5.9
2	2014	655.24	803.87	129.96	673.91	22.7	2.9
3	2015	565.75	746.40	134.80	611.60	31.9	8.1
4	2016	615.84	808.77	128.71	680.06	31.3	10.4
5	2017	652.73	813.79	140.68	673.11	24.7	3.1
Average		**620.70**	**792.59**	**134.78**	**657.81**	**27.7**	**6.0**

[a] Revenue of turbining for HPP System is equal to the net revenue of HPP System since there is no pumping, thus no associated cost.
[b] Net revenue = Revenue of turbining − Cost of pumping.

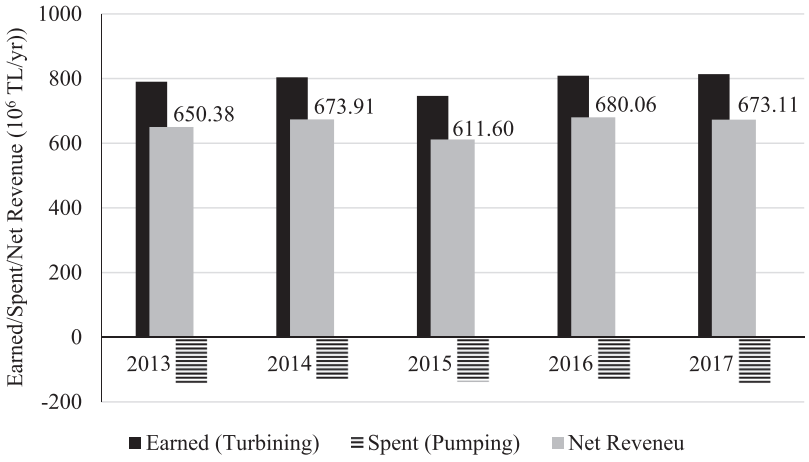


Fig. 8. Money earned from turbining, spend for pumping and net revenue obtained for PSH system.

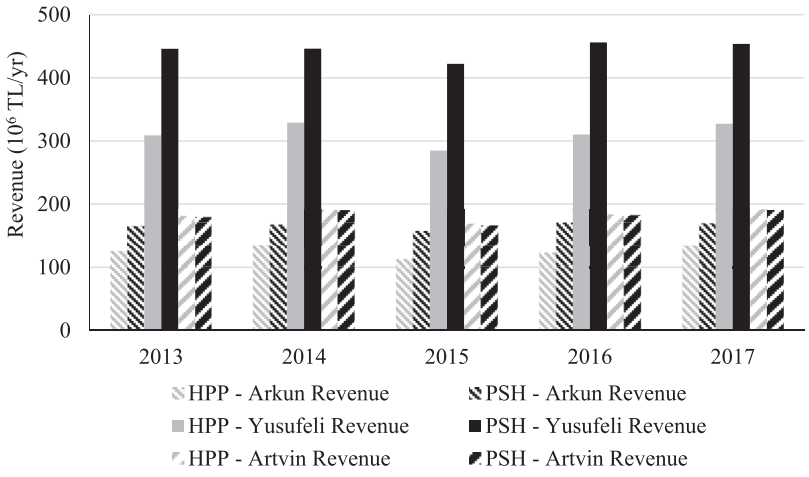


Fig. 9. Distribution of the revenue among three reservoirs for HPP and PSH systems.

converted to the PSH system. This holds for each of the individual years as well. The reason is that there is no pumping to Artvin reservoir since it lies downstream of the other two reservoirs in the cascade layout. Since there is no pumping, the revenue through turbining during the day decreases for Artvin reservoir while the other two reservoirs are able to increase their revenue due to turbining. Yusufeli, the middle reservoir, is the main contributor for the revenue gain. This is due to Yusufeli having an installed capacity of 540 MW and an active storage volume of 1464 hm^3, which are much larger than those of Arkun (Table 1). The large active storage allows higher amounts to be turbined and pumped, which results in higher revenue.

As Fig. 8 shows, converting the HPP into PSH results in a revenue gain. To assess the feasibility of such a conversion, we compare the annual revenue gain with the cost of conversion. For the case under study, reversible turbines (which are capable of both turbining and

pumping) need to be installed at Arkun and Yusufeli dams. The electro-mechanical equipment cost of a reversible turbine is a function of installed capacity, *IC* and net head, h_{net} and according to Ref. [42] can be approximated as follows:

$$Cost\left(in\ €x10^6\right) = 13.39\ (IC\ in\ MW)^{0.5825} \times (h_{net} in\ m)^{-0.3359} \quad (15)$$

Following the expression, cost of reversible turbine is calculated as 52.79 $€x10^6$ for Arkun and 88.25 $€x10^6$ for Yusufeli (see Table 1 for the installed capacity and net head values of the dams). Currently there are already conventional turbines installed. The cost of reversible turbine is approximately 20% higher than that of conventional turbine (as stated in Ref. [21]). Thus the required additional investment in turbines is estimated in total as 23.5 $€x10^6$. To make this cost value compatible with the annual revenue gain, an equivalent annuity (in the form of cost) is obtained through the suitable capital recovery factor (CRF). The annual inflation rate in Turkey is taken as 8.96% (averaged over years 2013–2017, The Central Bank of Republic of Turkey, https://www.tcmb.gov.tr), while the economic life of the turbine is taken as 50 years. The corresponding CRF is approximately 0.09, which results in an annuity of 2.11 $€x10^6$ for the additional cost of the turbines. Since annual gain is estimated as 12.04 $€x10^6$ (assuming $1€ = 2.53TL$ in 2013 with 15% annual increase in parity since then), for the case study under consideration, conversion is feasible.

Revenue gains calculated in this study (Table 2) are only due to operating the existing HPP system as a PSH system. Thus, it is assumed that all the energy needed to operate reversible turbines in the pumping mode is bought from the grid. In line with current energy policies of the country, the contribution of renewable energy sources such as wind in its energy budget is expected to increase. This will lower electricity prices especially for periods when the demand is low and energy generation from wind turbines is high. As a result the revenue gains will increase.

Finally, comparing the revenues across years (scenarios), it is observed that the change in the electricity price oscillations within a day may have a significant effect on the average annual revenue. Hourly electricity prices for 2013, 2014, 2015, 2016 and 2017 are given in Fig. 10. To make the comparison easy, x-axis of Fig. 10 represents the hours of the corresponding year (a total of 8760 h in a year) while the y-axis is the realized electricity prices for each of the 8760 h within a year. In Table 2, observe that 2016 electricity prices resulted in the largest percent net revenue gain whereas 2014 resulted in the lowest. This seems reasonable since 2016 has the largest daily variation with many outliers, whereas 2014 seems to have the lowest daily price variation (Fig. 10). One may expect that as the daily price variation increases so does the revenue, however since there are other factors that affect the gain such as how the daily price range changes from one month to the next, or the head of the reservoir, there may not be a linear or one-to-one correspondence between the range and the net revenue.

5.2. Operating rule curves for conventional and pumped-storage systems

One advantage of the proposed approach is that the optimum operation strategies (i.e. the operating rule curves for all the reservoirs in the cascade system) are derived simultaneously while estimating the annual net revenue of the system. In other words, solutions of the non-linear optimization problems in Sections 3.1 and 3.2 provide the maximum net revenues together with the operating rule curves for each hydropower plant that will result in those revenues.

The rule curves for Arkun, Yusufeli and Artvin for year 2013 are given in Figs. 11–13, respectively. The closest year to the average case is year 2013 (with an estimated benefit of 5.9%) and therefore we report the rule curves for that year. For the other four scenarios, operating rule curves for both HPP and PSH systems are obtained from the mathematical models as well. The operating rule curves for different scenarios (i.e. years 2013–2017) follow the same

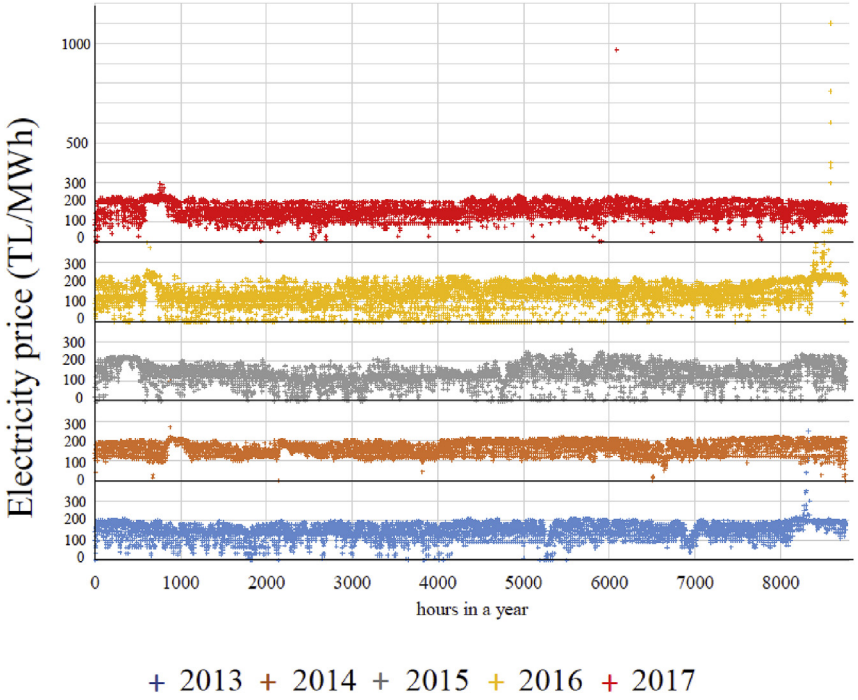


Fig. 10. Hourly electricity prices throughout the year (for years 2013–2017) [40].

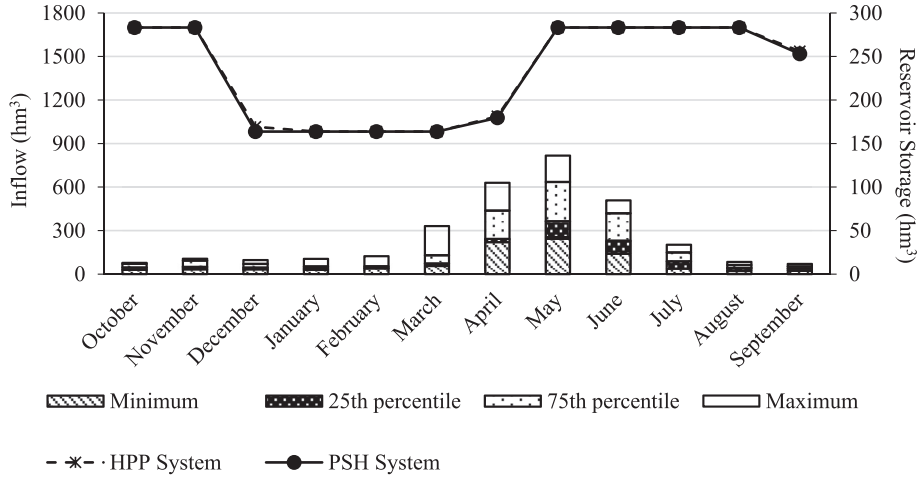


Fig. 11. Operating rule curves for Arkun HPP for 2013: Bars are for the Inflow (hm^3) and lines are for the Reservoir Storage (hm^3).

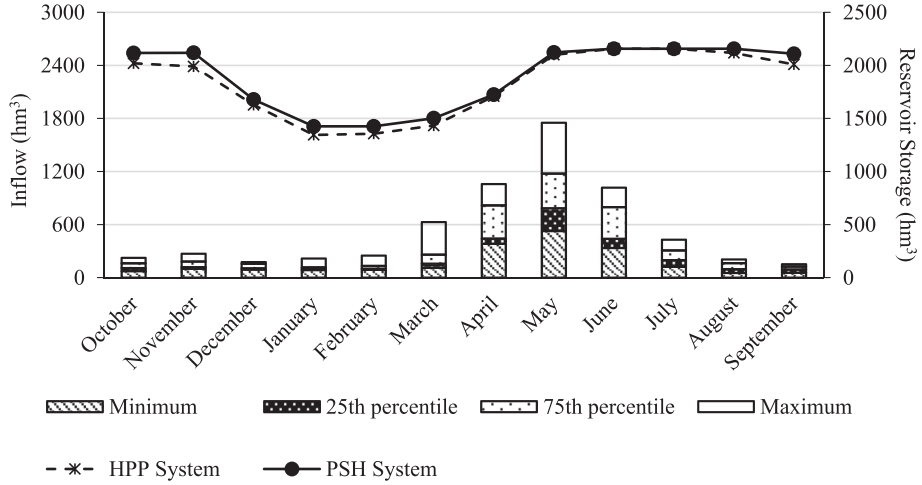


Fig. 12. Operating rule curves for Yusufeli HPP for 2013: Bars are for the Inflow (hm^3) and lines are for the Reservoir Storage (hm^3).

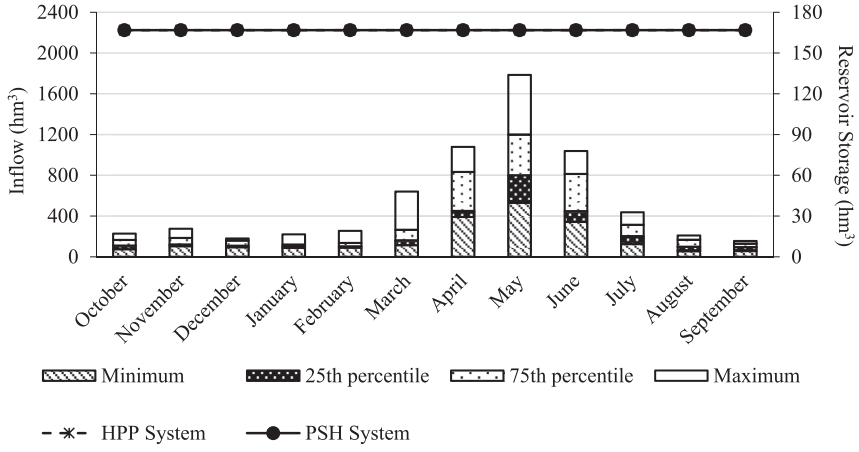


Fig. 13. Operating rule curves for Artvin HPP for 2013: Bars are for the Inflow (hm^3) and lines are for the Reservoir Storage (hm^3).

general trend with minor variations.

Based on the rule curves for PSH and HPP systems for each of the scenarios, we make several observations. First, for Artvin reservoir (the most downstream reservoir) the suggested end-of-month levels change neither across months nor across scenarios. The active storage volume of the reservoir of Artvin is relatively small and the head is much more important for energy generation. Therefore, it makes sense that the reservoir level is kept at its

maximum level over all scenarios. The rule curves are constant over the months for Artvin (Fig. 13). Second, for Arkun and Yusufeli reservoirs (i.e. reservoirs where pumping from the downstream reservoir is allowed), the changes in prices from one year to the other may affect the end-of-month reservoir levels, while the general trend is the same: During February, March, and April significant amount of water is turbined, and the storage levels are kept below the maximum storage levels. This is to account for the above-average inflows during May and June. During May–July period, reservoir levels are kept at maximum. Then the storage levels are lowered to benefit from the high electricity prices during August, November, and December (Figs. 11 and 12). Under pumped-storage, the amount of water turbined during high price months are not that high since the water is pumped during the night and the price difference during day and night gains importance.

Third observation is that going from HPP to PSH the rule curves change only very slightly. This implies, while the net turbined amount almost stays the same under PSH and HPP, the hours of the day during which water is turbined and pumped significantly affects the revenue generated. We would like to note that the only exception to closeness of rule curves under PSH and HPP is for months November–January for Arkun reservoir for year 2014. For those months of year 2014, the difference in end-of-month reservoir levels are more pronounced.

Our last observation is that the rule curves under PSH almost always lie above that of HPP. In PSH the turbined amount is gained back through pumping, and this slightly lowers down the net amount turbined.

5.3. *Operation of turbines in conventional and reversible turbines in pumped-storage mode*

We analyzed monthly average number of turbining and idle hours for the HPP system and pumping, turbining and idle hours for the PSH system for the rule curves given in Figs. 11–13. Since Artvin reservoir is the most downstream one, there is no pumping to it. Therefore total available time within a month is distributed between turbining and idle hours both for HPP and PSH systems as shown in Fig. 14a. We do not observe much of a change for Artvin reservoir. The percentage of time the Artvin's turbine works is 37.5% under HPP and 36.5% under PSH, respectively. The situation changes for Arkun and Yusufeli reservoirs since their turbines are replaced by reversible turbines when the HPP system is converted to the PSH system. While for the HPP system, the total available time within a month is distributed between turbining and idle hours, it is distributed between pumping, turbining and idle hours for the PSH system. Comparison of the monthly average number of pumping, turbining and idle hours for Arkun and Yusufeli reservoirs are given in Fig. 14b and c. Under the conventional HPP mode, Arkun's turbine works 43.5% of the time while Yusufeli's works 40.3% of the time, whereas under PSH, their reversible turbines work for 71.2% (57% turbining + 14.2% pumping) and 77.8% (58.7% turbining + 19.1% pumping) of the time, respectively. The figures indicate that for Arkun and Yusufeli, the percentage of time that energy is generated increases from 43.5% to 57% for Arkun, and increases from 40.3% to 58.7% for Yusufeli. Moreover, under PSH, the percentage of time that the turbine is used (for turbining or pumping) almost doubles compared to that of HPP.

We secondly observe that in accordance with these percentages, the amount of energy generated during the peak hours, i.e., when the energy is much needed, also increases for Arkun and Yusufeli. For Arkun the amount of energy generated per year increases from 758.1 GWh to 934.7 GWh, by 23.3%, and for Yusufeli, the amount of energy generated per year increases from 1794 GWh to 2451.6 GWh, by 36.7%. However, when the net energy amounts per

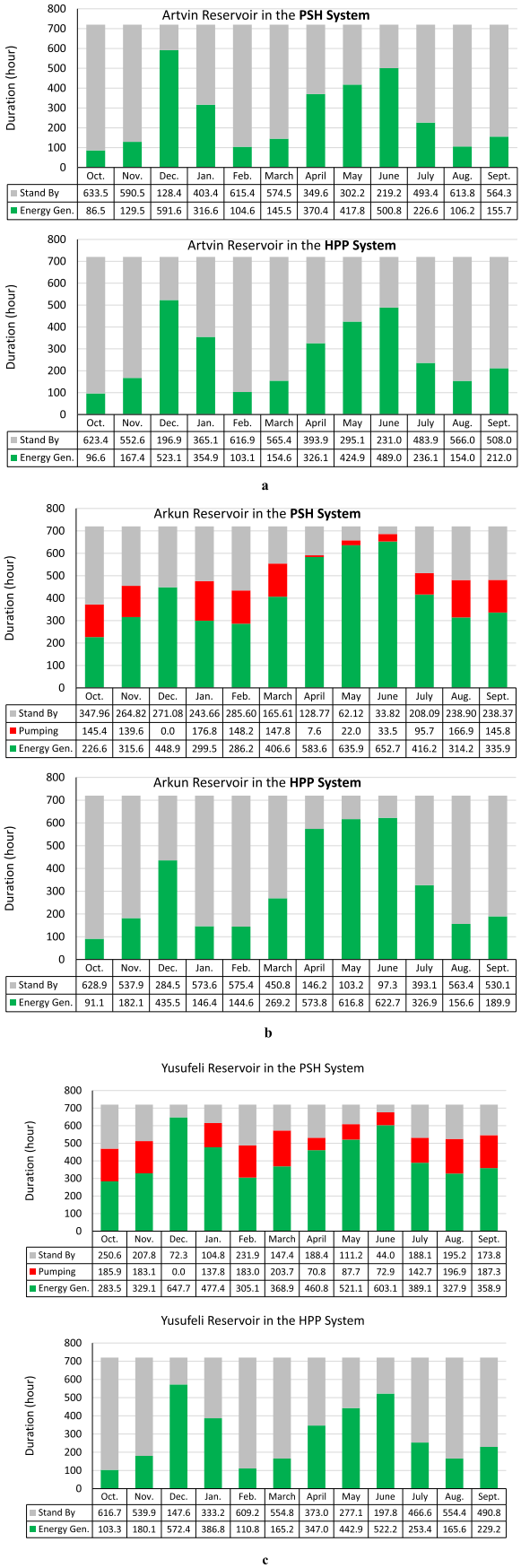


Fig. 14. a. Average number of turbining and idle hours for Artvin Reservoir for HPP and PSH systems (2013). b. Average number of pumping, turbining and idle hours for Arkun Reservoir for HPP and PSH systems (2013). c. Average number of pumping, turbining and idle hours for Yusufeli Reservoir for HPP and PSH systems (2013).

year are compared we observe a decrease as the system converted from conventional HPP to PSH. Under HPP, the total energy generated is 3592.2 GWh per year whereas under PSH, it is 3403.6 GWh per year. This amounts to a decrease of 5.2%.

6. Conclusions

Integration of renewable energy resources into a country's energy mix, especially for energy importing countries like Turkey, is very important to sustain energy security. Effective utilization of intermittent renewable energy resources, such as wind and solar, necessitates appropriate energy storage methods. Currently, PSH is the most viable large-scale energy storage alternative. Operating an already existing cascade hydropower plant as a PSH system is a more preferred alternative than building the PSH from scratch due to difficulties in site selection and environmental impacts of building new reservoirs. If operated optimally, PSH systems may provide additional monetary benefits due to hourly electricity price variations. In this study, optimal operation strategies for already existing cascade hydropower plants and for the proposed PSH system are developed considering the stochastic behavior of inflows and hourly electricity prices. To obtain realistic estimates of additional monetary benefits due to operation of existing cascade hydropower plants as a PSH system, hourly electricity price variations are appropriately represented in the optimization model following a scenario-based approach. Revenue gain due to operation of a cascade multi-reservoir system composed of three HPPs, namely Arkun, Yusufeli and Artvin located in the Coruh Basin of Turkey are calculated and compared in the presence and absence of pumped-storage.

A cascade system with three consecutive reservoirs is used to demonstrate application of the nonlinear mathematical model developed in this study. Although optimal solutions are obtained for this case study, as the number of consecutive reservoirs increase, the dimensionality may become a problem and the solver may have trouble in finding the optimal solution. Then one may resort to soft computing approaches, which we leave as a future work. We used a number of simplifications and assumptions in the proposed model: (i) fixed environmental flows are used, (ii) evaporation from and precipitation over the reservoir surfaces are assumed to be negligible in comparison with inflows, (iii) tail water elevations are assumed to be constant and (iv) all the electricity generated by the cascade system is sold.

It is demonstrated that when long-term operation strategies (i.e. operating rule curves) are devised using the optimization model proposed in this study, an average of 37 million TL (approximately 12.0 million €) additional revenue (i.e. from 620.70×10^6 TL/year for HPP system to 657.81×10^6 TL/year for PSH system) is secured for Çoruh system annually. When the analysis is conducted over different price scenarios the benefits lie between 18.67 million TL to 64.22 million TL (approximately $5.0 million € to $19.2 million €). Similar analysis can be conducted for the other potential existing HPP Systems together with benefit-cost analysis, where the cost is mainly due to installation of pumping system in prioritizing the HPP Systems to be converted into PSH Systems. When the rule curves under HPP and PSH are contrasted, it is observed that similar amounts of water are turbined monthly under both systems, however PSH alters the hours of the day during which water is turbined and pumped. This change in timing of the turbining together with pumping may result in significant gain in revenue. Similar analysis can be conducted for other potential cascade HPP systems in the country and the results may provide guidance in prioritizing the HPP systems to be converted to PSH systems.

Integration of renewable energy sources into the power system of Turkey is an obligation to enhance energy security and is listed among the primary energy policies ad priorities of the Ministry of Energy and Natural Resources [43]. Assessment of potential economic benefits of hybrid wind-PSH or solar-PSH systems is the topic of future research. Obtaining some of the necessary energy for pumping the water from the lower reservoir of the PSH system to the upper reservoir from wind or solar power plants instead of buying it from the grid will further increase the economic benefits.

Acknowledgements

The monthly streamflow data of two streamgages (E23A037 and E23A035) from 1993 to 2011 were bought from the General Directorate of State Hydraulic Works of Turkey. Daily electricity prices for 2013, 2014, 2015, 2016 and 2017 were obtained from Energy Markets Management Inc. of Turkey. This data is publicly available at https://rapor.epias.com.tr/rapor/xhtml/dgpGunlukRapor.xhtml [40].

References

[1] A. Harby, J. Sauterleute, M. Korpås, A. Killingtweit, E. Solvang, T. Nielsen, Pumped storage hydropower, in: Detlef Stolten, Viktor Scherer (Eds.), Transition to Renewable Energy Systems, Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim, Germany, 2013.
[2] IRENA (International Renewable Energy Agency), Renewable Energy Technologies: Cost Analysis Series. Volume 1: Power Sector, Issue 3/5, 2012 hydropower. Working paper. Available at, www.irena.org/Publications.
[3] C.K. Ekman, S.H. Jensen, Prospects for large scale electricity storage in Denmark, Energy Convers. Manag. 51 (6) (2010) 1140–1147.
[4] D. Connolly, M. Leahy, A Review of Energy Storage Technologies: for the Integration of Fluctuating Renewable Energy, Version 4.0, University of Limerick, Limerick, 2010.
[5] N. Sivakumar, D. Das, N.P. Padhy, A.R.S. Kumar, N. Bisoyi, Status of pumped hydro-storage schemes and its future in India, Renew. Sustain. Energy Rev. 19 (2013) 208–213.
[6] N. Löhndorf, D. Wozabal, S. Minner, Optimizing trading decisions for hydro storage systems using approximate dual dynamic programming, Oper. Res. 61 (4) (2017) 810–823.
[7] M. Gimeno-Gutiérez, R. Lacal-Arántegui, Assessment of the European Potential for Pumped Hydropower Energy Storage. A GIS-Based Assessment of Pumped Hydropower Storage Potential. JRC Scientific and Policy Reports, Report EUR 25940EN, Publication Office of the European Union, Luxembourg, 2013.
[8] A. Killingtveit, Hydropower, in: Detlef Stolten, Viktor Scherer (Eds.), Transition to Renewable Energy Systems, Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim, Germany, 2013.
[9] E. Pujades, P. Orban, S. Bodeuz, P. Archambeau, S. Erpicum, A. Dassargues, Underground pumped storage hydropower plants using open pit Mines: how do groundwater exchange influence the efficiency? Appl. Energy 190 (2017) 135–146.
[10] H. Ibrahim, A. Ilinca, J. Perron, Energy storage systems – characteristics and comparisons, Renew. Sustain. Energy Rev. 12 (5) (2008) 1221–1250.
[11] SHW (State hydraulic works of Turkey), Sources of Energy – Hydroelectric Energy, 2017. http://en.dsi.gov.tr/hizmet-alanlari. (Accessed 21 July 2017).
[12] D. Connolly, H. Lund, P. Finn, B.V. Mathiesen, M. Leahy, Practical operation strategies for pumped hydroelectric energy storage (PHES) utilizing electricity price arbitrage, Energy Policy 39 (7) (2011) 4189–4196.
[13] I.A.G. Wilson, E. Barbour, T. Ketelaer, W. Kuckshinrichs, An analysis of storage revenues from the time-shifting of electrical energy in Germany and Great Britain from 2010 to 2016, Journal of energy storage 17 (2018) 446–456.
[14] N. Lu, J.H. Chow, A.A. Desrochers, Pumped-storage hydro-turbine bidding strategies in a competitive electricity market, IEEE Trans. Power Syst. 19 (2) (2004) 834–841.
[15] T. Muche, Optimal operation and forecasting policy for pumped storage plants in day-ahead markets, Appl. Energy 113 (2014) 1089–1099.
[16] E.I. Tică, B. Popa, R. Popa, Annual performance estimation of a multireservoir system including pumped storage plant for the mean hydrological year, J. Energy Eng. 143 (6) (2017) 1–10, 04017058.
[17] J.I. Pérez-Díaz, R. Millán, D. García, I. Guisández, J.R. Wilhelmi, Contribution of re-regulation reservoirs considering pumping capability to environmentally friendly hydropower operation 48 (2012) 144–152.
[18] Z. Bregar, Short-term optimization of the new Avce pumping plant and three existing hydro power plants on the Soca river in Slovenia, Electr. Power Syst. Res. 77 (2007) 1410–1417.
[19] J.D. Hunt, M.A.V. Freitas, A.O. Pereira Junior, Enhanced-Pumped-Storage: combining pumped-storage in a yearly storage cycle with dams in cascade in Brazil, Energy 78 (2014) 513–523.
[20] J.I. Pérez-Díaz, M. Chazarra, J. García-González, C. Cavazzini, A. Stoppato, Trends and challenges in the operation of pumped-storage hydropower

plants, Renew. Sustain. Energy Rev. 44 (2015) 767–784.
[21] S.-E. Fleten, D. Haugstvedt, J.A. Steinsbø, M. Belsnes, F. Fleischmann, Bidding hydropower generation: integrating short- and long-term scheduling, in: Power System Computation Conference (PSCC 2011 Stockholm), Stockholm, Sweden, seventeenth ed., 2011.
[22] O.B. Fosso, M.M. Belsnes, Short-term hydro scheduling in a liberalized power system, in: International Conference on Power System Technology-POWERCON, Singapore; 21-24 Nov 2004, 2004.
[23] G. Pritchard, A.B. Philpott, P.J. Neame, Hydroelectric reservoir optimization in a pool market, Math. Program. 103 (3) (2005) 445–461.
[24] A. Borghetti, C. Ambrosio, A. Lodi, S. Martello, An MILP approach for short-term hydro scheduling and unit commitment with Head-Dependent reservoir, IEEE Trans. Power Syst. 23 (3) (2008) 1115–1124.
[25] O.B. Bozorg-Haddad, P.S. Ashofteh, S. Rasoulzadeh-Gharibdousti, M.A. Mariño, Optimization model for design-operation of pumped-storage and hydropower systems, J. Energy Eng. 140 (2) (2014) 1–11.
[26] Z.-K. Feng, W.-J. Niu, C.-T. Cheng, J. Zhou, Peak shaving operation of hydro-thermal-nuclear plants serving multiple power grids by linear programming, Energy 135 (2017) 210–219.
[27] M.M. Belsnes, O. Wolfgang, T. Follestad, E.K. Aasgård, Applying successive linear programming for stochastic short-term hydropower optimization, Electr. Power Syst. Res. 130 (2016) 167–180.
[28] M. Ak, E. Kentel, S. Savasaneril, Operating policies for energy generation and revenue management in single-reservoir hydropower systems, Renew. Sustain. Energy Rev. 78 (2017) 1253–1261.
[29] M.T.L. Barros, F.T.-C. Tsai, S.-I. Yang, J.E.G. Lopes, W.W.-G. Yeh, Optimization of large-scale hydropower system operations, J. Water Resour. Plann. Manag. 129 (3) (2003) 178–188.
[30] C.-T. Cheng, W.-C. Wang, D.-M. Xu, K.W. Chau, Optimizing hydropower reservoir operation using hybrid genetic algorithm and chaos, Water Resour. Manag. 22 (2008) 895–909.
[31] Z.-K. Feng, W.-J. Niu, J.-Z. Zhou, C.-T. Cheng, H. Qin, Z.-Q. Jiang, Parallel multi-objective genetic algorithm for short-term economic environmental hydro-thermal scheduling, Energies 10 (2) (2017) 163.
[32] M.B. Djukanović, M.S. Ćalović, B.V. Vešović, D.J. Šobajić, Neuro-fuzzy controller of low head hydropower plants using adaptive-network based fuzzy inference system, IEEE Trans. Energy Convers. 12 (4) (1997) 375–381.
[33] Z.-K. Feng, W.-J. Niu, C.-T. Cheng, X.-Y. Wu, Optimization of hydropower system operation by uniform dynamic programming for dimensionality reduction, Energy 134 (1) (2017) 718–730.
[34] B. Liu, C. Cheng, S. Wang, S. Liao, K.-W. Chau, X. Wu, W. Li, Parallel chance-constrained dynamic programming for cascade hydropower system operation, Energy 165 (2018) 752–767.
[35] Z.-K. Feng, W.-J. Niu, C.-T. Cheng, J.R. Lund, Optimizing hydropower reservoir operation via orthogonal progressive optimality algorithm, J. Water Resour. Plann. Manag. 144 (3) (2018), 04018001.
[36] Y. Shang, S. Lu, Y. Ye, R. Liu, L. Shang, C. Liu, X. Meng, X. Li, Q. Fan, China' energy-water nexus: hydropower generation potential of joint operation of the Three Gorges and Qingjiang cascade reservoirs, Energy 142 (2018) 14–32.
[37] M. Chazarra, J. García-González, J.I. Pérez-Díaz, M. Arteseros, Stochastic Optimization model for the weekly scheduling of a hydropower system in day-ahead and secondary regulation reserve markets, Electr. Power Syst. Res. 130 (2016) 67–77.
[38] M. Chazarra, J.I. Pérez-Díaz, J. Garcia-González, Optimal joint energy and secondary regulation reserve hourly scheduling of variable speed pumped storage hydropower plants, IEEE Trans. Power Syst. 33 (1) (2017) 103–105.
[39] M. Ak, Rule Curves for Operating Single- and Multi-Reservoir Hydropower Plants (Ph.D. thesis), Civil Engineering Department, Middle East Technical University, Ankara, Turkey, 2017.
[40] EPIAS (Energy Markets Management Inc.) Daily Prices, 2018. https://rapor.epias.com.tr/rapor/xhtml/dgpGunlukRapor.xhtml. (Accessed 17 June 2018).
[41] DOLSAR Engineering Company, Yusufeli Dam and HPP Feasibility Report, Ankara, 2010.
[42] G. AlZohbi, The cost of electromechanical equipment in a small hydro power storage plant, Journal of Energy Systems 2 (4) (2018) 238–259.
[43] MENR (Ministry of Energy and Natural Resources of T.C.), 2015-2019 Stratejik Planı (Strategic Plan). T.C. Enerji Ve Tabii Kaynaklar Bakanlığı, November 2017. https://sp.enerji.gov.tr/ETKB_2015_2019_Stratejik_Plani.pdf (in Turkish).